FINAL PROJECT · DATA ANALYSIS · JCBDAAH-005

Who to Save First: Reducing Home-Internet Churn at SambungHome

Identifying high-risk customers and designing a retention program that can be tested

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Purwadhika Digital Technology School · October 2026

26.6%
of 7,032 customers have churned

55.2%
churn among new month-to-month customers



SHIELD
4-layer retention program, proven first in a 3-month pilot

SambungHome is a fictional company. Customer data: IBM Telco Customer Churn (public dataset), n = 7,032 after cleaning.

EXECUTIVE SUMMARY

One in four customers leaves, mostly early; SambungHome SHIELD targets the four drivers behind it

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Considerations

Home broadband is set to grow ≈ **2.4×** by 2029 (21% → 50% of households)

All customers

26.6%

New month-to-month

55.2%

Winning a new customer costs **5–25×** more than keeping one

Key Question

Which customers should SambungHome target with retention offers, and when?

Building a retention program that is specific and can be tested.

Strategy

SambungHome SHIELD · 4 layers

Layer 1
Commitment Pack
1-year plan for new M2M

Layer 2
Auto-Pay Switch
for e-check payers

Layer 3
Protection Pack
3-month free add-ons

Layer 4
Risk Priority List
High Risk first

Impacts

< 26.6%
overall churn in 12 months

55.2% → 42.7%
new M2M toward benchmark

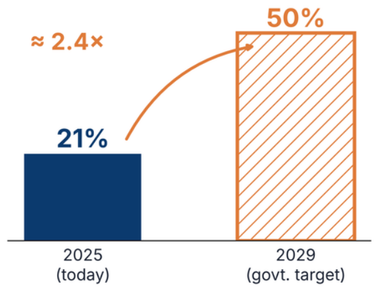
1,012
High Risk contacted in 30 days

Protecting the first six months is how SambungHome turns market growth into customer growth.

Indonesia's home-internet market is set to more than double, but growth only counts if customers stay

📈 Market opportunity

Households with fixed broadband



Internet users



229.4 M people, mostly via mobile

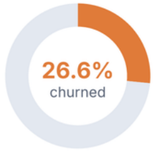
5–25× **more expensive to win a new customer than to keep one**

Growth only pays off if customers stay.

🏠 SambungHome's challenge

What SambungHome sells (fictional provider)

- **Home internet** — Fiber optic or DSL
- **Home phone** — single or multiple lines
- **6 paid add-ons** — security, backup, tech support…
- **Monthly billing** — month-to-month, 1- or 2-year



1,869 of 7,032 customers have already left

❓ Key question

Which customers should SambungHome target with retention offers, and when?

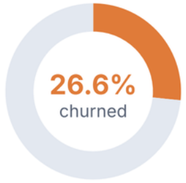
💡 KEY INSIGHT Home broadband is set to grow about **2.4×** by 2029, but SambungHome already loses **1 in 4 customers**. Every customer who leaves must be replaced at **5–25×** the cost of keeping them.

Source: APJII 2025 via ANTARA; Komdigi via Bisnis.com (2025); Gallo, HBR (2014); IBM Telco dataset (n = 7,032)

PROBLEM STATEMENT

26.6% of SambungHome customers have churned, and the company cannot yet tell who will leave next

📉 What we see



1,869

of **7,032** customers have left: about **1 in 4**

1 percentage point of churn ≈ **70 customers**

They leave early: median time with SambungHome



Churned	10 months
Stayed	38 months

❓ What SambungHome cannot answer yet



When is a customer most at risk?
Early months, or later in the relationship? → RQ1–2



What separates leavers from stayers?
Contract, payment, services, demographics? → RQ3



Who should be contacted first?
No priority list for the retention team → RQ4



MAIN QUESTION **Which customers should SambungHome target with retention offers, and when?**



RESULT TODAY

Retention offers go to everyone equally: **costly**, **not targeted** and **not measurable**, while every leaver must be replaced at **5–25×** the cost of keeping them.

Source: IBM Telco Customer Churn dataset (n = 7,032 after cleaning); Gallo, HBR (2014)

Four research questions, each tied to one goal and one business decision

Before building the model, we need to understand the questions behind it.

	Method	So that…
RQ1: How big is the churn problem, and do churned customers differ from those who stay in tenure and billing tier?	Churn rate · Mann-Whitney U	We have a baseline for every target
RQ2: Do new customers (0–6 months) churn more, and how does contract type change churn at each tenure stage?	Chi-square · Cramér's V	We know who gets a contract offer, and when
RQ3: Which of 18 attributes are linked to churn, and do e-check payment and missing security/tech support still raise churn within the same contract?	Ranking by Cramér's V	We act only on attributes that matter and can be changed
RQ4: Can the strongest attributes be combined into one risk score, and what share of churn do high-risk customers account for?	6-flag risk score	The retention team has a priority list

Each question uses a different variable and test, so the four answers do not overlap, and each one feeds one goal on the next slides.

The Customer Retention Manager decides; the analysis covers all 7,032 customers but not revenue

Stakeholders

Primary: Customer Retention Manager

Decides which customers receive which offer, and is accountable for the churn KPI.

Supporting

Customer Service contacts customers · **Finance** approves offer costs · **Marketing** designs the offers.



Scope

In scope

- All 7,032 customers; 16 categorical + 3 numeric attributes
- Statistical tests, rule-based risk score, benchmark-based targets

Out of scope

- Proving cause and effect; ML prediction model
- Network quality, complaints, usage (not in data)
- Revenue figures: billing period not defined in source



Dataset

7,032

customers after cleaning (7,043 raw × 21 columns)

1 row = 1 customer. One snapshot, no dates.

IBM Telco Customer Churn, a public sample dataset.



Source: IBM Telco Customer Churn dataset; mentor consultation (revenue excluded)

DATA DICTIONARY

7.043
Customers

21
Columns

Service Characteristics	
PhoneService	Whether the customer subscribes to phone service
MultipleLines	Whether the customer has multiple phone lines
InternetService	Type of internet service subscribed
OnlineSecurity	Whether the customer subscribes to online security service
OnlineBackup	Whether the customer subscribes to online backup service
DeviceProtection	Whether the customer subscribes to device protection service
TechSupport	Whether the customer subscribes to technical support service
StreamingTV	Whether the customer subscribes to TV streaming service
StreamingMovies	Whether the customer subscribes to movie streaming service

Target Variable	
Churn	Customer churn status (Yes = Churn, No = Retained)

Customer Account	
tenure	Number of months the customer has subscribed
Contract	Customer contract type
PaperlessBilling	Whether paperless billing is enabled
PaymentMethod	Customer payment method
MonthlyCharges	Monthly subscription charge
TotalCharges	Total charges accumulated during the subscription period

Customer Demographics	
customerID	Unique customer identifier
gender	Customer gender
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Indicates whether the customer has a partner
Dependents	Indicates whether the customer has dependents

Source: IBM Telco Customer Churn dataset; mentor consultation (revenue excluded)

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The cleaned data holds 7,032 customers with no duplicates or outliers, ready for analysis

21 columns in four groups

4
Demographics
gender, senior, partner, dependents

7
Account
ID, tenure, contract, billing, payment, charges

9
Services
phone, internet and 6 add-ons

1
Target
Churn (Yes / No)

Cleaning and preparation

7,043 rows
raw data, 21 columns

→

− 11 rows
blank TotalCharges, all tenure 0, none churned (0.16%)

→

0 found
duplicates; outliers by IQR in 3 numeric columns

→

7,032 rows
+ 4 engineered features (ChurnFlag, TenureGroup, ChargeGroup, SeniorLabel)

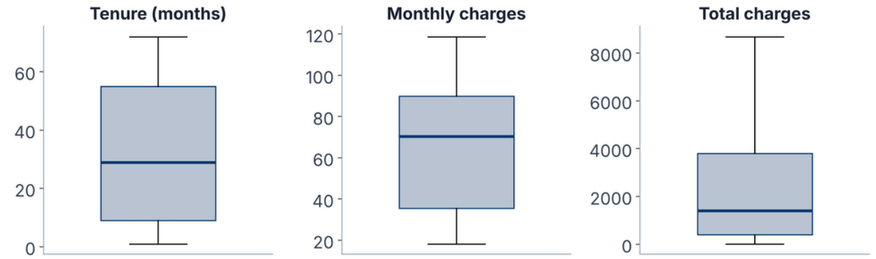
REVISION MonthlyChargeGroup uses the same three tiers for every customer. The old version split tiers inside each internet type, so “Low” fiber cost more than “High” DSL.

Source: IBM Telco Customer Churn dataset; analysis notebook sections 4–5

No outliers in any numeric column, and the boxplots already show churners are newer and pay more

① All customers: IQR check

Column	Q1	Q3	Lower	Upper	Outliers
tenure	9	55	−60	124	0
MonthlyCharges	35.59	89.86	−45.82	171.27	0
TotalCharges	401.45	3,794.74	−4,688.48	8,884.67	0



② Split by churn status



⏳ **Tenure:** churned median **10** vs stayed **38** months

💳 **Monthly charges:** churned median **79.65** vs **64.45**

🧾 **Total charges:** churned lower (703.55 vs 1,683.60) because they leave early

KEY INSIGHT No outliers by IQR, so all 7,032 rows are kept. The dots in the churned group are long-tenure customers who still left: real customers, not data errors.

Source: Analysis notebook section 5.3 (IQR method: Q1 − 1.5×IQR and Q3 + 1.5×IQR)

ANALYTICAL APPROACH

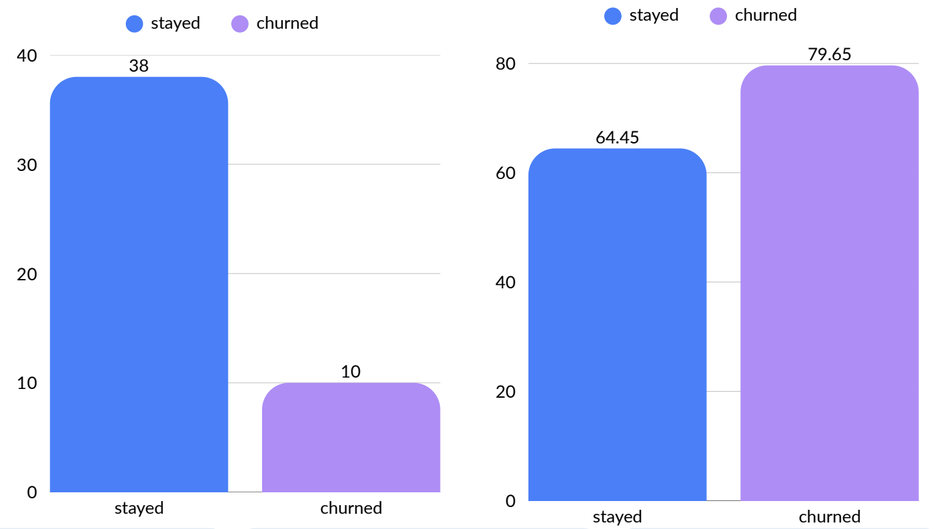
Each method was chosen to fit the data type, with a 5% significance level

Method	Why this method	Used in
Churn rate, share of churners	Describes the size of each group	RQ1–RQ4
Chi-square test of independence	Churn and the attributes are categorical	RQ2–RQ4
Cramér's V	Chi-square says whether; V says how strong (0.1 weak · 0.3 moderate · 0.5 strong)	RQ2–RQ4
Mann-Whitney U (+ Shapiro-Wilk)	Tenure and charges are not normally distributed (Shapiro $p < 0.05$)	RQ1
Rule-based risk score	Combines the strongest flags into one explainable list	RQ4

SIGNIFICANCE $\alpha = 0.05$: a wrong conclusion costs one retention campaign, not safety or a large investment. Every key result has $p < 0.001$, so it holds even at $\alpha = 0.01$.

Churned customers are far newer than those who stay; the billing difference is real but small

Churned vs stayed: medians



26.6%
churn rate

10 vs 38
median tenure in months, churned vs stayed

≈ 70
customers per point

① Medians and normality (Shapiro-Wilk)

Variable	Churned	Stayed	Shapiro (churned)	Shapiro (stayed)
tenure	10	38	1.5e-23	4.3e-16
MonthlyCharges	79.65	64.45	1.6e-13	4.4e-17

All Shapiro p < 0.05 → not normal → use Mann-Whitney U

② Mann-Whitney U test

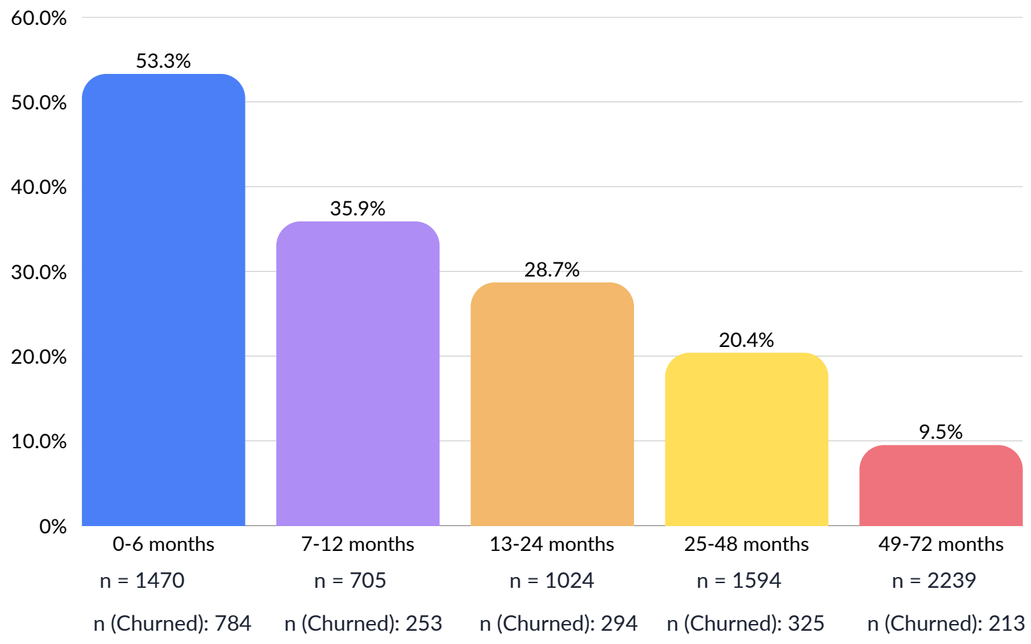
Variable	p-value	Effect size r	Strength	Result
tenure	6.0e-211	0.48	medium	Reject H0
MonthlyCharges	8.5e-54	0.24	small	Reject H0

ANSWER TO RQ1 Churners are much newer customers; their billing tier is higher but the effect is small. Tenure is the lead to follow in RQ2.

Source: IBM Telco Customer Churn dataset (n = 7,032); Mann-Whitney U with rank-biserial effect size r

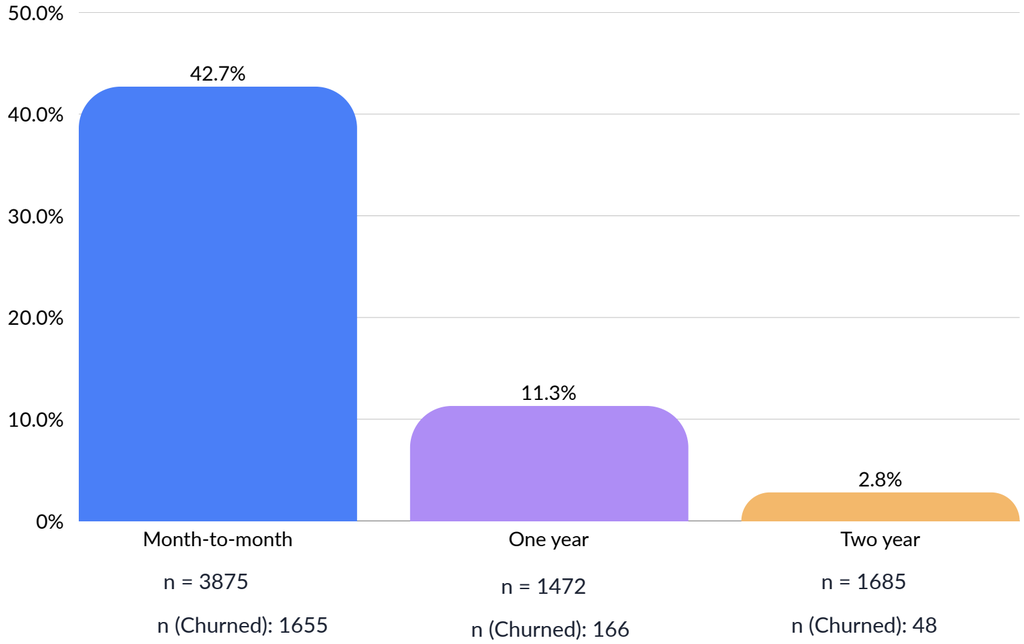
Churn peaks in the first six months and among customers without a contract

Churn rate by tenure (months)



0–6 months: **53.3%** churn, **41.9%** of all churners · V = 0.37

Churn rate by contract



Month-to-month: **42.7%** churn, **88.6%** of all churners · V = 0.41

KEY INSIGHT Churn falls at every tenure stage and is 15× higher on month-to-month than on 2-year contracts (both χ^2 $p < 0.001$).

Source: IBM Telco Customer Churn dataset (n = 7,032); chi-square test, Cramér's V

RQ2 · RISKIEST SEGMENT

New month-to-month customers are the riskiest segment: 55.2% churn and 41.7% of all churners

Churn rate: tenure × contract

	Month-to-month	One year	Two year
0–6 months	55.2% n=1,413	10.5% n=38	0.0% n=19
7–12 months	42.0% n=581	10.6% n=85	0.0% n=39
13–24 months	37.7% n=737	8.1% n=197	0.0% n=90
25–48 months	32.9% n=802	10.6% n=518	2.2% n=274
49–72 months	26.0% n=342	12.9% n=634	3.3% n=1,263

What the heatmap shows

Contract matters at every stage

Month-to-month 26–55%; 1-year 8–13%; 2-year 0–3%

Tenure matters within the same contract

Among month-to-month only: χ^2 $p < 0.001$, $V = 0.21$

Together they are stronger

Tenure × contract $V = 0.45$ vs 0.37 and 0.41 alone

Small cells are not used for targets

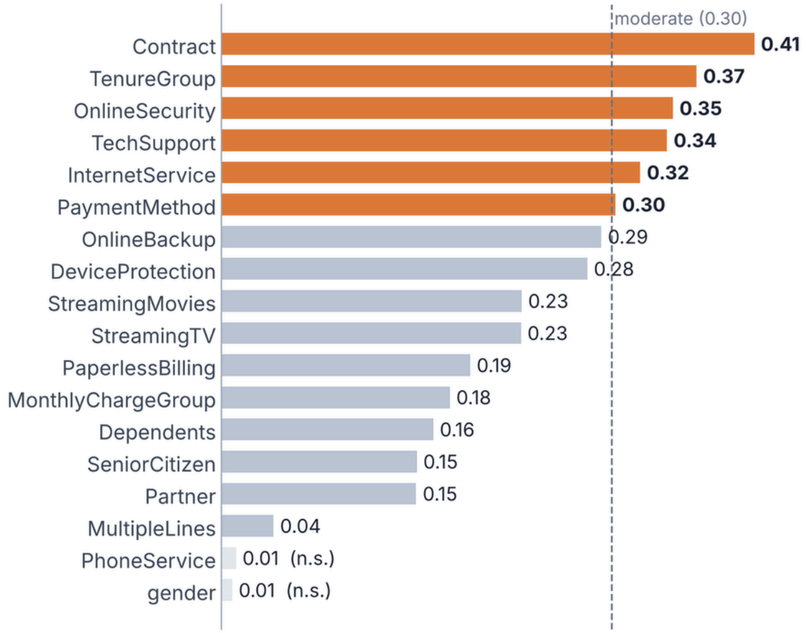
e.g. 2-year at 0–6 months has n = 19

ACTION Offer a 1-year contract upgrade to month-to-month customers, starting in their first six months (Layer 1 · Commitment Pack → G2).

Source: IBM Telco Customer Churn dataset (n = 7,032); n = customers in each cell

Six of 18 attributes have a moderate link with churn; gender and phone service have none

Strength of link with churn (Cramér's V)



Covered in RQ2

Contract (0.41), **Tenure** (0.37)

Strongest links overall

Actionable

Online Security (0.35), **Tech Support** (0.34), **Payment method** (0.30)

The company can change these

Context only

Internet type (0.32): fiber 41.9% vs DSL 19.0%

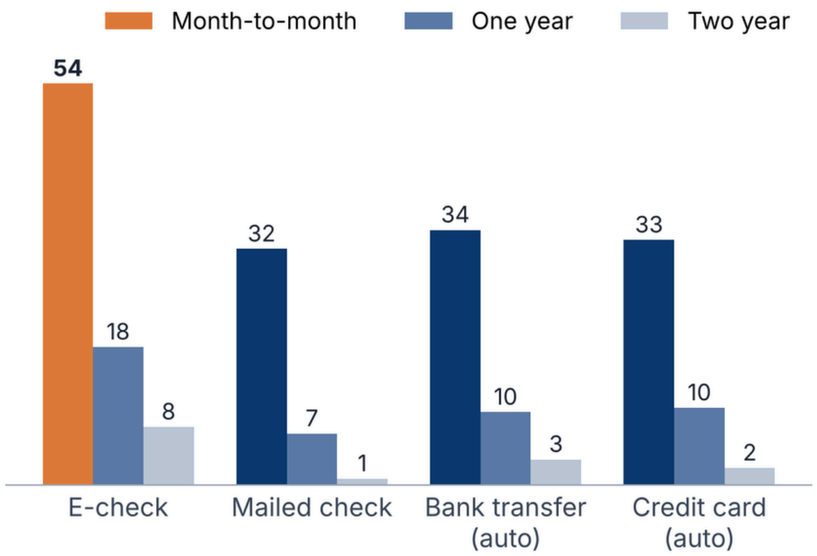
No network or complaint data to explain why; 92.5% of the High charge tier is fiber

KEY INSIGHT Demographics are weak (V ≤ 0.16) and gender and phone service are not significant (p > 0.3), so they are not used for targeting.

Source: IBM Telco Customer Churn dataset (n = 7,032); chi-square test and Cramér's V for each attribute

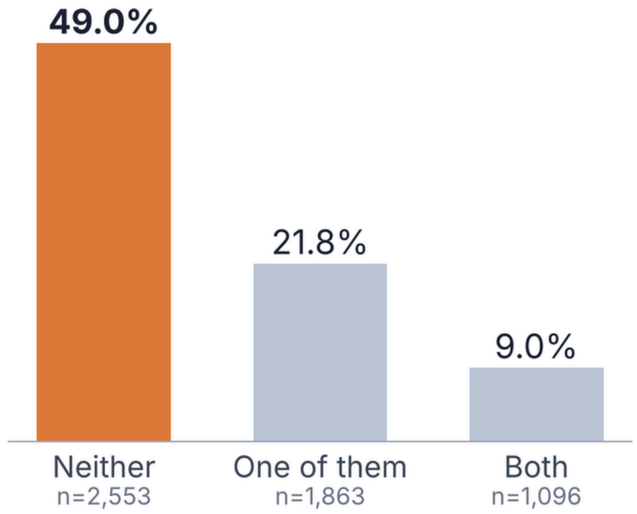
E-check payers and customers without security or tech support churn about twice as often

Churn by payment method and contract (%)



Within month-to-month: e-check **53.7%** vs 31.6–34.1% · V = 0.21, p < 0.001

Internet users: security + tech support



Neither: **49.0%** vs both: **9.0%** · V = 0.36

ACTION Auto-pay switch for e-check payers (Layer 2 → G3a) and a free 3-month Security + Tech Support trial (Layer 3 → G3b).

Source: IBM Telco Customer Churn dataset; payment chart n = 7,032, add-on chart n = 5,512 internet users

RQ4 · WHO TO CONTACT FIRST

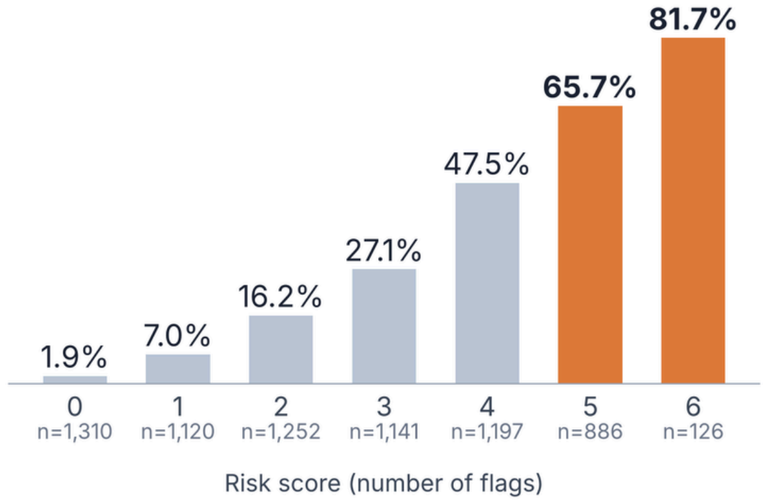
A six-flag risk score isolates 1,012 customers who churn at 67.7%

Six flags, +1 point each

Tenure 0–6 months | Month-to-month | E-check payment | High charge tier | No Online Security | No Tech Support

Segment	Customers	Churn	Share of churners
High (5–6 flags)	1,012 (14.4%)	67.7%	36.7%
Medium (3–4)	2,338 (33.2%)	37.6%	47.0%
Low (0–2)	3,682 (52.4%)	8.3%	16.4%

Churn rate by risk score



Risk score V = **0.51** (strong), above any single attribute (max 0.41)

KEY INSIGHT High Risk profile: median tenure 4 months, 100% month-to-month, 96.5% e-check, ~98% without security or support. High + Medium = 47.6% of customers but 83.6% of churners.

Source: IBM Telco Customer Churn dataset (n = 7,032); flags chosen from RQ2–RQ3

KEY FINDINGS

Churn is concentrated in new, uncommitted customers who pay manually and have no add-ons

RQ	Answer	Evidence
RQ1	1 in 4 customers churned (26.6%); churners are much newer (median 10 vs 38 months)	Mann-Whitney $p < 0.001$, $r = 0.48$
RQ2	New month-to-month customers churn 55.2% and make up 41.7% of all churners	χ^2 $p < 0.001$, $V = 0.45$
RQ3	E-check (53.7% within month-to-month) and no security/tech support (49.0% vs 9.0%) are the changeable drivers	χ^2 $p < 0.001$, $V = 0.21–0.36$
RQ4	1,012 High Risk customers churn at 67.7% and account for 36.7% of churners	Risk score $V = 0.51$

SO WHAT Churn is concentrated, not random: the same small group is new, uncommitted, paying manually and unprotected, so a few targeted offers can reach most of the risk.

SambungHome SHIELD: four layers of protection against churn, each tied to one driver and one goal

LAYER 1 Commitment Pack

Fixes
No contract commitment

Who
New month-to-month customers (0–6 months) · **1,413**

Goal G2 · churn
55.2% ➡ 42.7%
toward all month-to-month

LAYER 2 Auto-Pay Switch

Fixes
Manual monthly payment

Who
E-check payers on month-to-month · **1,850**

Goal G3a · churn
53.7% ➡ ≈ 33%
toward auto-pay, same contract

LAYER 3 Protection Starter Pack

Fixes
No protective add-ons

Who
Internet users without Security & Tech Support · **2,553**

Goal G3b · churn
49.0% ➡ 21.8%
toward one add-on

LAYER 4 Risk Priority List

Fixes
No priority for outreach

Who
High Risk customers (score 5–6) · **1,012**

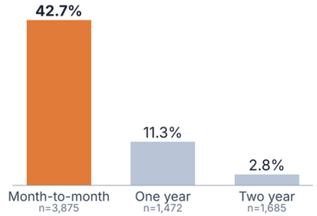
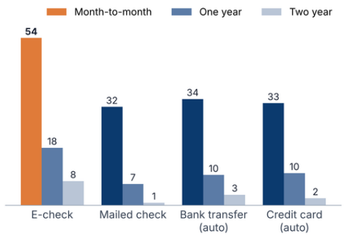
Goal G4 · churn
67.7% ➡ 37.6%
toward Medium Risk

G1 · OVERALL

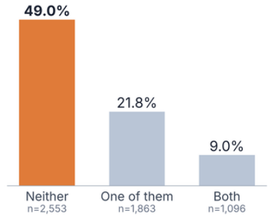
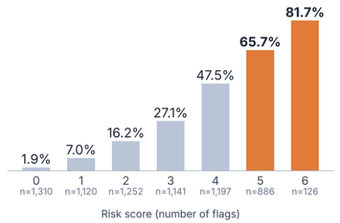
Overall churn falls **below 26.6%** in 12 months. Each target is an **internal benchmark**: a better-performing group in our own data. The real effect is measured in a 3-month pilot.

Source: Baselines and benchmarks from RQ2–RQ4 (analysis notebook sections 7–9); no projection model

Commit and automate: lock in new customers early and remove the monthly decision to pay manually

	① Why it matters	② The offer	③ How it works
LAYER 1 Commitment Pack	 New month-to-month churn **55.2%** = **41.7%** of all churners	**1-year contract upgrade** **15% off** the monthly bill for 12 months + **free installation** 🎯 **Churn 55.2% → toward 42.7%**	🔧 **Day 0:** technician offers it at installation 📞 **Day 25:** CS calls before the first bill 📅 **Month 6:** last chance, offer closes ⏱️ *Valid in the first 6 months; reviewed at month 3*
LAYER 2 Auto-Pay Switch	 E-check **53.7%** vs auto-pay **32.8–34.1%**, same contract	**Switch e-check to auto-pay** **5% off** the monthly bill for 6 months when auto-pay is set up 🎯 **Churn 53.7% → toward ≈ 33%**	🔧 **Day 0:** auto-pay set up at installation 📱 **Day −5:** app/SMS reminder with sign-up link 🔁 **Every bill:** repeat for the first 3 months ⏱️ *Sent every billing cycle for 3 months*

Protect and prioritise: give new customers a support safety net and call the riskiest first

	① Why it matters	② The offer	③ How it works
LAYER 3 — Protection Starter Pack	 No add-ons **49.0%** vs both **9.0%**; **71.2%** of internet churners	**Free Online Security + Tech Support** **3 months free**, active from installation, plus technician check-in calls 🎯 **Churn 49.0% → toward 21.8%**	🔧 **Day 0:** technician activates both add-ons 📞 **Day 7 & 30:** check-in calls from Tech Support 💬 **Week 11:** offer to continue as paid add-on ⏱️ *3-month trial, then paid continuation*
LAYER 4 — Risk Priority List	 High Risk: **14.4%** of customers, churn **67.7%**	**Monthly call list from the risk score** Retention team calls **High Risk first**, then Medium, and offers Layers 1–3 as each customer qualifies 🎯 **1,012 called in 30 days; churn → toward 37.6%**	🧮 **Day 1:** score every customer (6 flags, +1 each) 📋 **Week 1:** High Risk list to the retention team 🔄 **Monthly:** refresh scores and the list ⏱️ *Refreshed at the start of every month*

SHIELD follows a new customer from installation day through the critical first six months

Installation
Day 0

What happens
First contact with the technician; contract and payment are chosen here

⚠ Risk in our data
Default = riskiest path: no contract, manual pay

SHIELD in action
- L1 offer
- L2 set-up
- L3 activated

First 30 days
Day 1–30

What happens
First experience of speed and support; first problems appear

⚠ Risk in our data
Month 1 churn **62.0%**, the highest of the year

SHIELD in action
- L3 check-in day 7
- L4 list

First bill
Day 25–30

What happens
First manual payment: a natural moment to reconsider

⚠ Risk in our data
Month 2 churn still **51.7%**

SHIELD in action
- L1 CS call
- L2 reminder

Months 2–6
Critical window

What happens
Habit forms, or the customer leaves

⚠ Risk in our data
0–6 months: **53.3%** churn, **41.9%** of churners

SHIELD in action
- L1 closes M6
- L3 paid offer
- L4 monthly

WHY THE JOURNEY MATTERS The first 30 days decide most of the churn, so SHIELD starts at installation, not after a customer shows signs of leaving.

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A staged 12-month rollout: prove it on High Risk first, then scale only what works

	Months 1–3 · Pilot	Month 3	Months 4–6 · Expand	Months 7–12 · Full launch
🔒 Commitment Pack	High Risk new month-to-month	Go / adjust	Add Medium Risk	Standard offer at every installation
💳 Auto-Pay Switch	High Risk e-check payers	Go / adjust	Add Medium Risk	Auto-pay set up at every installation
🛡️ Protection Pack	High Risk internet users	Go / adjust	Add Medium Risk	Trial bundled with every new plan
🎯 Risk Priority List	Score all; split offered vs comparison	Go / adjust	Monthly refresh, call Medium	Score + KPIs in the dashboard

Layer	Possible risk	Mitigation
🔒 Commitment Pack	Discount cost is higher than the value of customers kept	Size the discount with Finance; test it in the pilot first
💳 Auto-Pay Switch	Customers distrust automatic debit	Send a reminder before every debit; easy opt-out
🛡️ Protection Pack	Trial users do not continue as paid	Check-in calls on day 7 and 30; paid offer at week 11
🎯 Risk Priority List	Offers work less well than the data suggests	Month-3 checkpoint against the comparison group before scaling

MONTH-3 CHECKPOINT Continue only if the offered group churns less than the comparison group; otherwise adjust the offer before scaling.

CONCLUSION

Churn is concentrated and predictable, so SHIELD targets each driver in the first months

🎯 Research Question	🔍 Finding	🧭 Strategy	📈 Outcome
RQ1 · G1 How big is churn, and who leaves?	• 1 in 4 churned: **26.6%** (1,869 of 7,032) • Churners are new: **10 vs 38** mo tenure	🗓️ **Act early** Every layer starts at installation	**<26.6%** overall (G1) within 12 months
RQ2 · G2 New customers & contract type?	• New month-to-month churn **55.2%** • …and they are **41.7%** of all churners	🔒 **L1 · Commitment Pack** 1-year plan, 15% off, months 0–6	**42.7%** new M2M (G2) target, from 55.2%
RQ3 · G3 Which attributes can we change?	• E-check (M2M) churn **53.7%** • No add-ons **49.0%** vs both add-ons 9.0%	💳 **L2 Auto-Pay** · 🛡️ **L3 Protect** Auto-pay incentive · free add-on trial	**33%** e-check (G3a) add-ons 21.8% (G3b)
RQ4 · G4 Can one score rank who to call?	• 6-flag risk score: **V = 0.51**, strongest • High Risk (14.4%) = **36.7%** of churners	🎯 **L4 · Risk Priority List** Monthly risk score as the call list	**1,012** called in 30 days High Risk (G4)

✅ **In total** **SambungHome SHIELD** moves each high-risk group toward a better-performing group in our own data, and the 3-month pilot with a comparison group measures the real effect.

Each limitation has a concrete next step, most of them answered by the pilot

Limitation	Next step
Public sample data, not company CRM data	Re-run the notebook on SambungHome's own customer data
Billing period of MonthlyCharges not defined → no revenue figures	Add revenue impact once internal billing data is available
Associations, not cause and effect	Test through the pilot with a comparison group
Size of each offer's effect is unknown; offer costs unknown	Measure it in the pilot; cost the offers with Finance
No network, complaint or usage data; fiber churn unexplained	Add these data and analyse fiber customers separately
Single snapshot; equal-weight rule-based score	Refresh monthly; re-check the flags with pilot data

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Thank you

Save the first six months, secure the contract, automate the payment, and protect the connection.

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APPENDIX

Notebook: Remedy 4.ipynb (sections 1–12)

Dataset: IBM Telco Customer Churn (public)

Dashboard: [link]

References: APJII via ANTARA (2025); Komdigi via Bisnis.com (2025); Gallo, Harvard Business Review (2014)